

4. Measuring Yield

1. What is the effective annual yield if the semiannual periodic interest rate is 4.3%?

In our problem, the periodic interest rate is a semiannual rate of 4.3% and the frequency of payments is twice per year. Inserting these numbers, we have:

$$\text{effective annual yield} = (1.043)^2 - 1 = 1.087849 - 1 = 0.087849 \text{ or about } \mathbf{8.785\%}.$$

2. Suppose that the coupon rate of a floating-rate security resets every six months at a spread of 70 basis points over the reference rate. If the bond is trading at below par value, explain whether the discount margin is greater than or less than 70 basis points.

If the bond is trading below par value, then the discount margin or assumed annual spread (basis points) will be greater than 70 basis points. This is because the spread must increase to make the present value of the cash flows less than the par value.

3. An investor is considering the purchase of a 20-year 7% coupon bond selling for \$816 and a par value of \$1,000. The yield to maturity for this bond is 9%. Answer the below questions.

(a) What would be the total future dollars if this investor invested \$816 for 20 years earning 9% compounded semiannually?

To determine the future value of any sum of money invested today, we use the below equation: $P_n = P_0 (1 + r)^n$ where n = number of periods, P_n = future value n periods from now (in dollars), P_0 = original principal (in dollars), and r = interest rate per period (in decimal form). Inserting in our values, we have: $P_n = P_0(1 + r)^n = \$816(1.045)^{40} = \$816(5.8163645) = \mathbf{\$4,746.15}$.

(b) What are the total coupon payments over the life of this bond?

The total dollar amount of coupon interest is found by multiplying the semiannual coupon interest by the number of periods: total coupon interest = nC . Thus, the total coupon payments are: $nC = 40(\$35) = \mathbf{\$1,400.00}$.

(c) What would be the total future dollars from the coupon payments and the repayment of principal at the end of 20 years?

There are several ways to approach this problem. One method is to compute the present value of the cash flows and then multiply this by the future value factor for a lump sum. Another method (which involves less work) is to compute the future value of all the

cash flows. For this method, we would (i) compute the future value of the annuity cash flows which is the coupon interest plus interest on interest, and (ii) add the par value which occurs at maturity which is $M = \$1,000$. The equation is:

$$P_n = \text{coupon interest plus interest on interest} + M = C \left[\frac{(1+r)^n - 1}{r} \right] + M$$

where P_n is the future value of all cash flows at time N , C is the amount of the semiannual coupon annuity in dollars, r = annual interest rate / number of times interest paid per year (where we assume interest is reinvested at r), n = number of times interest paid per year times the number of years, and M = par value at the end of the period.

Using this formula and inserting our values, we have:

$$P_n = C \left[\frac{(1+r)^n - 1}{r} \right] + M = \$35 \left[\frac{(1.045)^{40} - 1}{0.045} \right] + \$1,000 = \$35[107.03032] + \$1,000 =$$

$$\$3,746.06 + \$1,000 = \mathbf{\$4,746.06}.$$

(d) For the bond to produce the same total future dollars as in part (a), how much must the interest on interest be?

We can note that the future value of the interest payment just computed in part (c) is \$3,746.06 and the coupon payments over the life of the bond computed in part (b) is \$1,400. The difference is the interest on interest, which is \$2,346.06.

Another way of computing the interest on interest is to note that it is the difference between the coupon interest plus interest on interest and the total dollar coupon interest, as expressed by the formula:

$$\text{interest on interest} = C \left[\frac{(1+r)^n - 1}{r} \right] - nC.$$

$$\text{Inserting in our values gives: } \$35 \left[\frac{(1.045)^{40} - 1}{0.045} \right] - 40(\$35) = \$35[107.03032] -$$

$$\$1,400 =$$

$$\$3,746.06 - \$1,400.00 = \mathbf{\$2,346.06}.$$

(e) Calculate the interest on interest from the bond assuming that the semiannual coupon payments can be reinvested at 4.5% every six months and demonstrate that the resulting amount is the same as in part (d).

Since the computation assumes interest on interest is invested at 4.5% we have the same computation given in part (d) where the yield to maturity of 4.5% was used in computation. Once again, we have:

$$\text{interest on interest} = C \left[\frac{(1+r)^n - 1}{r} \right] - nC$$

where r is still 4.5%. Inserting in our values gives: $\$35 \left[\frac{(1.045)^{40} - 1}{0.045} \right] - 40(\$35) =$

$\$35[107.03032] - \$1,400 = \$3,746.06 - \$1,400.00 = \mathbf{\$2,346.06}$ which is the same amount as in part (d).

4. What is the total return for a 20-year zero-coupon bond that is offering a yield to maturity of 8% if the bond is held to maturity?

For zero-coupon bonds, none of the bond's total dollar return is dependent on the interest-on-interest component, so a zero-coupon bond has zero reinvestment risk if held to maturity. The yield earned on a zero-coupon bond held to maturity is equal to the promised yield to maturity. This is because whenever one can reinvest the coupon payments at the yield to maturity, then the total return will be the same as the yield to maturity. Thus, the total return is **8%**.

5. Suppose that an investor with a five-year investment horizon is considering purchasing a seven-year 9% coupon bond selling at par. The investor expects that he can reinvest the coupon payments at an annual interest rate of 9.4% and that at the end of the investment horizon two-year bonds will be selling to offer a yield to maturity of 11.2%. What is the total return for this bond?

The investor has a five-year investment horizon to purchase a seven-year 9% coupon bond for \$1,000. The yield to maturity for this bond is 9% since it is selling at par. The investor expects to be able to reinvest the coupon interest payments at an annual interest rate of 9.4% and that at the end of the planned investment horizon the then-two-year bond will be selling to offer a yield to maturity of 11.2%. The total return for this bond is found as follows:

Step 1: Compute the total coupon payments plus the interest on interest, assuming an annual reinvestment rate of 9.4%, or 4.7% every six months. The coupon payments are \$45 every six months for five years or ten periods (the planned investment horizon). Applying equation (3.7), the total coupon interest plus interest on interest is

$$\text{coupon interest plus interest on interest} = C \left[\frac{(1+r)^n - 1}{r} \right] = \$45 \left[\frac{(1.047)^{10} - 1}{0.047} \right] =$$

$$\$45[12.40162]$$

$$= \$558.14.$$

Step 2: Determining the projected sale price at the end of five years, assuming that the required yield to maturity for two-year bonds is 11.2%, is accomplished by calculating the present value of four coupon payments of \$45 plus the present value of the maturity value of \$1,000, discounted at 5.6%. As seen below, the projected sale price is **\$961.53**.

projected sale price = present value of coupon payments + present value of par value =

$$C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] + \left[\frac{M}{(1+r)^n} \right] = \$45 \left[\frac{1 - \frac{1}{(1.056)^4}}{0.056} \right] + \left[\frac{\$1,000}{(1.056)^4} \right] =$$

$$\$45[3.4970813] + \$1,000[0.8041634] = \$157.37 + \$804.16 = \mathbf{\$961.53}.$$

Step 3: Adding the amounts in steps 1 and 2 gives total future dollars of \$558.14 + \$961.53 = **\$1,519.67**.

Step 4: To obtain the semiannual total return, compute the following:

$$\left[\frac{\text{total future dollars}}{\text{purchase price of bonds}} \right]^{1/h} - 1 = \left[\frac{\$1,519.67}{\$1,000} \right]^{1/10} - 1 = [1.63840]^{0.16667} - 1 =$$

$$1.042738 - 1 = 0.042738 \text{ or } \mathbf{4.2738\%}.$$

Step 5: Double 4.2738%, for a total return of about **8.55%**.